

Medication adherence is a critical factor in managing chronic illnesses, yet many patients forget to take their prescribed doses on time. This can lead to serious health complications, reduced treatment effectiveness and increased medical costs. Traditional pillboxes rely on manual tracking, which can be ineffective with memory challenges or busy schedules.

This project aims to develop a smart pillbox using an Arduino and an MC-38 magnetic sensor to monitor whether the pillbox has been opened within a specified timeframe. If the pillbox remains unopened past the scheduled medication time, the system will trigger an alert (such as a buzzer, LED, or mobile notification) to remind the user to take their medication. By automating adherence tracking, this smart pillbox can help patients manage their medication schedules more effectively, improving health outcomes and reducing the risk of missed doses.

Some negative aspects of the first attempt was that we created it too small. We underestimated the size of our hardware.